

LEVEL 1 – BASIC THEORETICAL QUESTIONS (BOARD LEVEL)

Focus: Definitions, simple explanations, examples.

Target: 150 questions (approx.)

1. Basics of Electrochemistry

1. Define electrochemistry.
 2. State the scope of electrochemistry.
 3. Give two examples of conductors and two examples of insulators.
 4. What is an electrolyte? Give an example.
 5. Differentiate between strong and weak electrolytes.
 6. Define ionic conductivity.
 7. List factors affecting the conductivity of electrolytes.
 8. How does temperature affect conductivity?
 9. How does concentration affect conductivity?
 10. Define mobility of ions.
 11. Explain the term molar conductivity.
 12. Write the unit of conductivity.
 13. Give one practical application of ionic conductivity.
 14. Explain why pure water is a poor conductor.
 15. What is meant by limiting molar conductivity?
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2. Electrolytic and Galvanic Cells

16. Define an electrolytic cell.
 17. Give one example of an electrolytic cell.
 18. Describe the construction of an electrolytic cell.
 19. Define a galvanic (voltaic) cell.
 20. Give one example of a galvanic cell.
 21. Write the standard cell notation for a Zn-Cu cell.
 22. List differences between electrolytic and galvanic cells.
 23. Explain the function of electrodes in a cell.
 24. What is the role of electrolyte in a cell?
 25. Define anode and cathode.
 26. Which electrode is positive in a galvanic cell?
 27. Which electrode is negative in an electrolytic cell?
 28. Give an industrial application of an electrolytic cell.
 29. Explain how a battery works in terms of galvanic cells.
 30. What is standard EMF of a cell?
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3. Electrode Potential

31. Define electrode potential.
 32. What is standard electrode potential (E°)?
 33. Name three reference electrodes.
 34. What is the potential of SHE?
 35. Write one advantage of calomel electrode.
 36. Write one advantage of Ag/AgCl electrode.
 37. State the Nernst equation.
 38. Give one application of the Nernst equation.
 39. How is cell potential calculated under non-standard conditions?
 40. Explain why a more positive electrode potential means a stronger oxidizing agent.
 41. Why is SHE considered standard?
 42. What is the meaning of reduction potential?
 43. How does concentration affect electrode potential?
 44. Give an example of a half-cell reaction.
 45. Explain the term "electrochemical series."
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4. Electrochemical Series

46. Define electrochemical series.
 47. State one significance of the electrochemical series.
 48. Predict the direction of a redox reaction using electrochemical series.
 49. Which metal corrodes fastest, Fe or Cu? Explain.
 50. List two uses of the electrochemical series.
 51. How can the electrochemical series predict reactivity of metals?
 52. Define oxidizing and reducing agents using E° .
 53. Explain why noble metals are less reactive.
 54. Predict if Zn will displace Cu from CuSO_4 .
 55. Explain corrosion tendency using electrochemical series.
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5. Faraday's Laws of Electrolysis

56. State Faraday's first law of electrolysis.
57. State Faraday's second law of electrolysis.
58. Define electrochemical equivalent.
59. Write the formula for calculating mass deposited during electrolysis.
60. Name two practical applications of electrolysis.
61. Explain electroplating.
62. Explain electrefining.
63. How is water decomposed by electrolysis?
64. Give one industrial application of Faraday's laws.

65. How is the quantity of electricity related to mass of substance deposited?

6. Types of Electrochemical Cells

66. Define primary cell.
 67. Give one example of a dry cell.
 68. Give one example of a Leclanché cell.
 69. Define secondary/rechargeable cell.
 70. Give one example of a lead-acid battery.
 71. Give one example of a Ni-Cd cell.
 72. Give one example of a Li-ion cell.
 73. What is a fuel cell?
 74. Give one example of a hydrogen-oxygen fuel cell.
 75. Give one example of an alkaline fuel cell.
 76. Mention one advantage of secondary cells over primary cells.
 77. State one disadvantage of fuel cells.
 78. Compare primary and secondary cells.
 79. Mention the use of hydrogen-oxygen fuel cell in space applications.
 80. Give one advantage of Li-ion cells over lead-acid cells.
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7. Corrosion

81. Define corrosion.
 82. List types of corrosion.
 83. Explain the electrochemical theory of corrosion.
 84. Give one method to prevent corrosion.
 85. Explain galvanization.
 86. Explain cathodic protection.
 87. Give one example of alloying to prevent corrosion.
 88. Why does iron rust faster than gold?
 89. What is the role of moisture in corrosion?
 90. Mention one industrial method to slow down corrosion.
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8. Batteries and Storage Devices

91. List two primary batteries.
92. List two secondary batteries.
93. Explain the working principle of a lead-acid battery.
94. Explain the working principle of a Ni-Cd battery.
95. Explain the working principle of a Li-ion battery.

96. Mention one application of primary batteries.
 97. Mention one application of secondary batteries.
 98. Give one advantage of batteries in portable devices.
 99. State one disadvantage of lead-acid batteries.
 100. State one disadvantage of Ni-Cd batteries.
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9. Conductometric, Potentiometric, and Voltammetric Methods

101. Define conductometry.
 102. Give one application of conductometry.
 103. Define potentiometry.
 104. Give one application of potentiometry.
 105. Define voltammetry.
 106. Give one analytical application of voltammetry.
 107. Explain how conductivity changes are measured.
 108. Explain how cell potential is measured potentiometrically.
 109. Give one advantage of voltammetry.
 110. Differentiate potentiometry and conductometry.
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10. Applications of Electrochemistry

111. Explain electroplating.
112. Explain electrorefining.
113. What is the chlor-alkali process?
114. Give one application of electrolysis of molten salts.
115. How is electrochemistry used to determine the concentration of solutions?
116. How is electrochemistry used in energy storage?
117. Give one industrial application of electroplating.
118. Give one industrial application of electrorefining.
119. Explain how fuel cells convert energy.
120. Mention one environmental application of electrochemistry.